SAKSHAM JUNEJA

AI/ML Undergraduate | Computer Vision | Real-Time AI Systems

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PROFESSIONAL SUMMARY

AI/ML undergraduate (B.Tech, expected 2028) focused on computer vision and real-time perception systems. Designed and implemented end-to-end machine learning pipelines spanning data engineering, model training, inference, and deployment., from data engineering and model training through inference and deployment: a five-stage real-time motion-recognition system running entirely on CPU, a CNN-based chess engine trained on 14M+ games, and a facial-recognition attendance platform in active production use by 500+ students. Seeking Machine Learning / Computer Vision internship roles.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, JavaScript, C, HTML/CSS

Machine Learning & Deep Learning: Convolutional Neural Networks (CNN), Transfer Learning, Model Training & Evaluation, Hyperparameter Tuning, Data Preprocessing, Feature Engineering

Computer Vision: Pose Estimation, Real-Time Video Inference, Face Recognition, Motion & Action Recognition

Frameworks & Libraries: TensorFlow, Keras, OpenCV, MediaPipe, face_recognition, NumPy, Pandas, Scikit-learn, Streamlit, Flask

Backend & Data: Firebase (Authentication, Realtime Database), REST APIs

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Linux

Core CS Foundations: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering Practices, Version Control

PROJECTS

TAI (Tekken AI) — Real-Time Motion-to-Combat Vision System **Jun 2026 – Present**

Python, OpenCV, MediaPipe, Streamlit

- Designed a five-stage real-time computer vision pipeline — MediaPipe-based pose estimation and velocity tracking, atomic action detection, temporal buffering, intent resolution, and move execution — that reconstructs multi-action motion sequences from a webcam feed and converts them into in-game combat commands.
- Mapped recognized motion sequences to a 62-move combo database through a dedicated resolver module, deliberately separating motion interpretation from move execution so new combos or games could be added without changing the perception stack.
- Optimized the full pipeline to run in real time on CPU-only hardware with a standard laptop webcam, trading a smaller supported action vocabulary for consistent low-latency response without GPU dependency.

CNN-Based Chess Engine — Learned Position Evaluation + Classical Search **Mar 2026 – May 2026**

Python, TensorFlow, Keras, ResNet, Minimax, Alpha-Beta Pruning

- Trained a ResNet-based CNN on 14M+ professional chess games to output a position-evaluation score, encoding each board as a 16-plane spatial tensor rather than a flat feature vector to preserve piece-position relationships.
- Built a chunked NPZ streaming pipeline to feed the training loop directly from disk in batches, avoiding the need to hold the full dataset in memory while keeping GPU utilization high during training.
- Combined the learned evaluator with classical game-tree search (Minimax with Alpha-Beta pruning, move ordering, transposition tables, and quiescence search) so the network approximated position value while exact tactics were resolved by search; tuned search depth and evaluation weighting using development-time games against Stockfish.

AI Attendance System — Real-Time Facial Recognition Platform **Jan 2025 – Mar 2025**

Python, OpenCV, face_recognition, Firebase, Streamlit

- Designed a Firebase-backed pipeline (Authentication + Realtime Database) so attendance records sync immediately across devices without a custom backend server.
- Split the system into two independently deployable Streamlit apps — a recognition engine and an admin monitoring dashboard — so the monitoring interface could be updated without touching the recognition pipeline.

EDUCATION

Bachelor of Technology (B.Tech), Artificial Intelligence & Machine Learning **Expected 2028**

Maharishi Dayanand University, Haryana, India

LEADERSHIP & EXPERIENCE

Technical Lead — Vaish College **Aug 2024 – Jul 2025**

- Led technical mentorship for students, running workshops on Python, Git/GitHub, Firebase, and ML fundamentals, and coordinating hackathons and technical events.

Campus Ambassador — Zuno by Foundit (Paid) & HackWithIndia **Aug 2024 – Present**

- Onboarded 100+ students as a paid campus ambassador for Zuno by Foundit; separately ran technical outreach for HackWithIndia to promote hackathon participation.